

Zuohao Cao

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EDUCATION

Wuhan University

Wuhan, China

B.S. in Computer Science and Technology

Sep 2024 – Jun 2028 (Expected)

GPA: 3.5/4.0

Core Courses: Advanced Mathematics, Linear Algebra, Probability and Mathematical Statistics, Data Structures, Algorithm Design and Analysis.

HONORS & AWARDS

- **First Prize**, “Huazhong Cup” Mathematical Modeling Challenge 2025
- **First Prize**, National University Student Internet of Things Design Competition (Central China Division) 2025
- **First Prize**, National College Student Computer System Capability Competition (Xiaomi Cup) 2025

RESEARCH & PROJECT EXPERIENCE

Mathematical Modeling and Data Analysis

2025

“Huazhong Cup” Mathematical Modeling Challenge | Core Programming Lead

- **Statistical Modeling:** Built mathematical models for complex systems; utilized Python (NumPy, SciPy, Pandas) for large-scale data preprocessing and feature extraction.
- **Algorithm Optimization:** Applied regression analysis and optimization algorithms to identify patterns in noisy datasets.
- **Data Visualization:** Used Matplotlib to plot complex model evolution trends, assisting the team with qualitative and quantitative interpretations of theoretical models.

Computer Vision & Robotics System

2025

National College Student Computer System Capability Competition (Xiaomi Cup) | Vision Algorithm Developer

- **Signal Processing:** Developed vision tracking and detection algorithms, focusing on noise filtering and target recognition in real-time data streams.
- **System Integration:** Implemented efficient communication protocols between software nodes to process high-dimensional real-time data streams.

Embedded Systems & Hardware Development

2025

National University Student IoT Design Competition (Central China Division) | Embedded Developer

- **Sensor Data Acquisition:** Designed multi-layer PCB layouts and wrote C++ firmware for real-time sensor data acquisition.
- **Cloud LLM Training:** Trained large models in the cloud to optimize delivery riders’ routes and developed physical AI products.

INTERNSHIP EXPERIENCE

Beijing Si-Tech Information Technology Co., Ltd.

2026

Software Engineering Intern

- **Data Stream Processing:** Learned and applied Apache Flink for high-throughput data stream management, gaining a deep understanding of system architecture design.
- **Agent Development:** Explored the development of an AI agent based on Large Language Models (LLMs) tailored for Wuhan tourism.

PROFESSIONAL SKILLS

- **Programming Languages:** Python (Proficient in Data Science/Modeling), C/C++ (Proficient in Embedded/Algorithms), SQL, MATLAB (Familiar).
- **Technical Tools:** ROS2, Git, Linux, PCB Design, Apache Flink.
- **Language Proficiency:** English (IELTS 6.0, CET-6 503). Strong academic literature reading and writing capabilities.